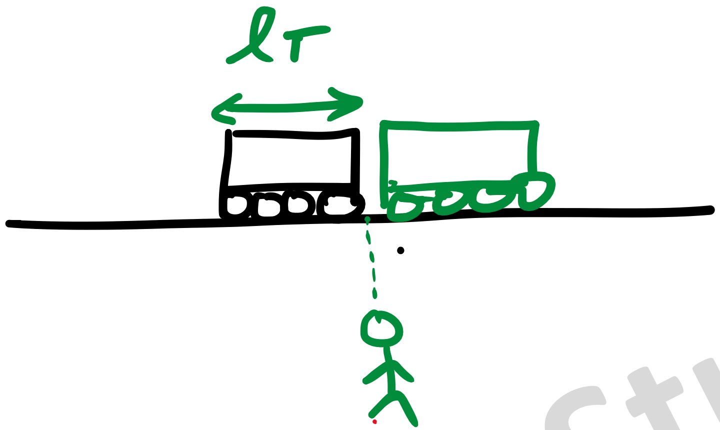
①



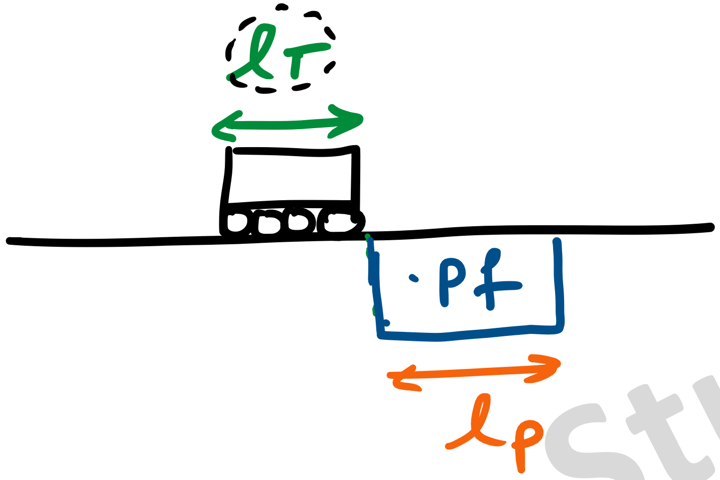
$$\boxed{S = \frac{D}{T}}$$

Time taken by train to cross the person

$$\Rightarrow \boxed{T = \frac{D}{S} = \frac{l_T}{v_T}}$$

②

$$\boxed{S = \frac{D}{T}}$$



Time taken by train to cross the platform

$$\Rightarrow T = \frac{D}{S} = \frac{l_p + l_T}{V_T}$$

Q] A train that is 280 metres long, travelling at a uniform speed, crosses a platform in 60 s and passes a man standing on the platform in 20 s . What is the length of the platform in metres?

[GATE 2014 EC]

1. $l_T = 280m$

2. Crosses a platform : $T_1 = 60s$

3. Crosses the man : $T_2 = 20s$

4. $l_p = ?$

Crosses a platform : $T_1 = 60s$

$$\frac{\cancel{l_T}^{280} + l_p}{V_T_{(14)}} = 60 \Rightarrow l_p = 60 \times 14 - 280 = 560m$$

Crosses the man : $T_2 = 20s$

$$\frac{\cancel{l_T}^{280m}}{V_T} = 20s$$

$$\boxed{V_T = 14\ m/s}$$

Q] It takes 10s and 15s respectively for two trains travelling at different constant speeds to completely pass a telegraph post. The length of the first train is 120m and that of the second train is 150m. The magnitude of the difference in the speeds of the two train (in m/s) is___

(A)2
(B)10
(C)12
(D)22
[GATE 2016 EC]

$l_{Telegraph} = l_{person} = l_{Tree} = 0$

① $\boxed{T = \frac{l_T}{V_T}}$

$10s = \frac{\cancel{l_{T_1}} \rightarrow 120\,(m)}{V_{T_1}}$

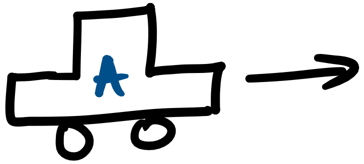
$V_{T_1} = 12\ m/s$

$15s = \frac{\cancel{l_{T_2}} \rightarrow 150m}{V_{T_2}}$

$V_{T_2} = 10\ m/s$

· Avg = 12 − 10 = 2 m/s //

Relative Speed



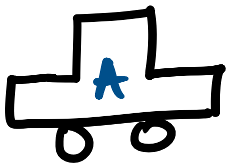
$10\ km/hr$, $1\ hr$, $d = ?$

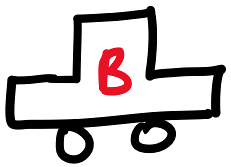
$$S = D/T \rightarrow D = ST$$

$$= 10\ km/hr \times 1\ hr$$

$$= 10\ km$$

//

 → 10 km/hr

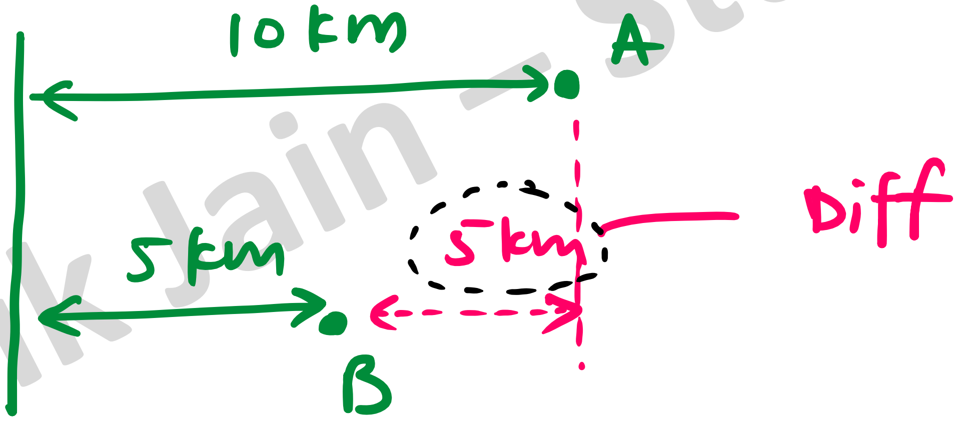
 → 5 km/hr

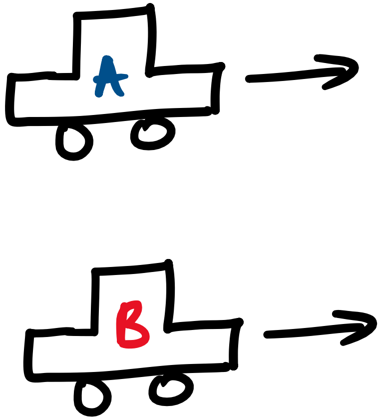
$$\boxed{D = ST}$$

Relative speed

$$S = \frac{D}{T} = \frac{5\text{ km}}{1\text{ hr}}$$

$$\boxed{S_r = 5\text{ km/hr}}$$





10 km/hr

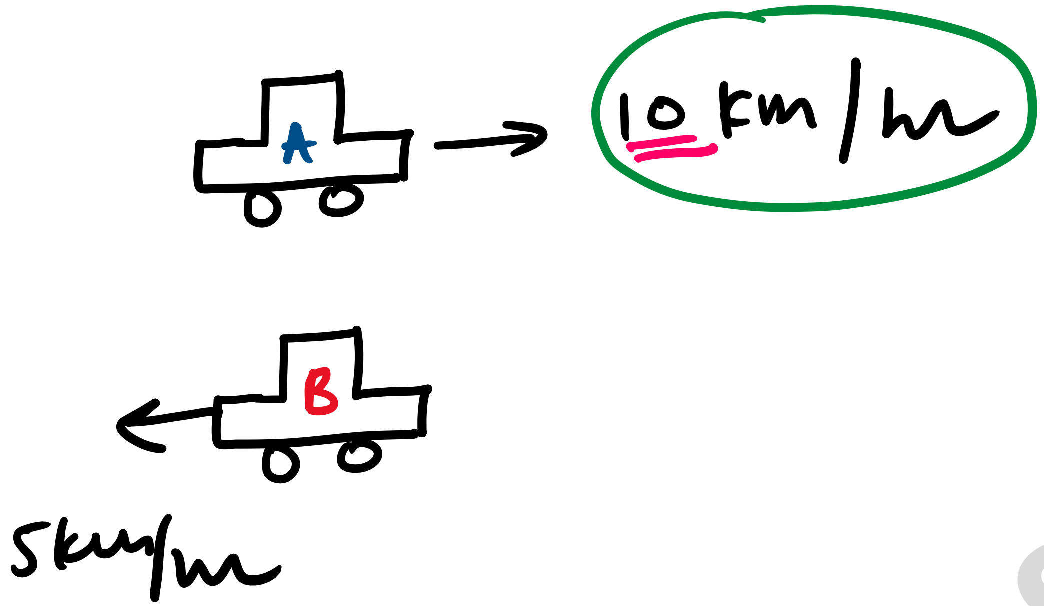
5 km/hr

(same)

$S_r = v_1 - v_2$ $(v_1 > v_2)$

$S_r = 10 - 5$

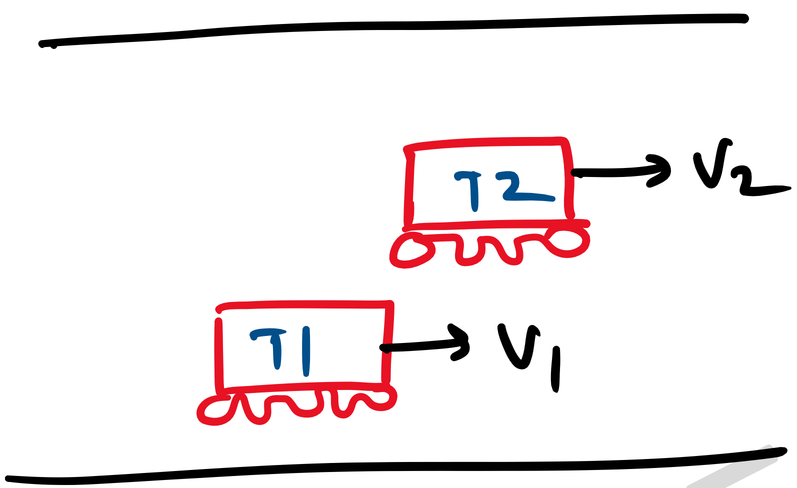
$= 5$ km/hr.



[opposite]

$S_r = v_1 + v_2$

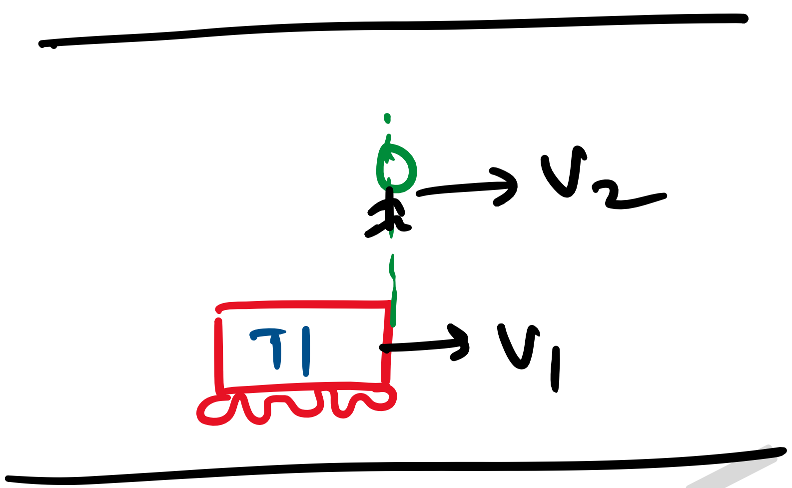
$S_r = 10 + 5 = 15$ km/hr.



Time taken to cross each other

$$\boxed{T = \frac{D}{S}}$$

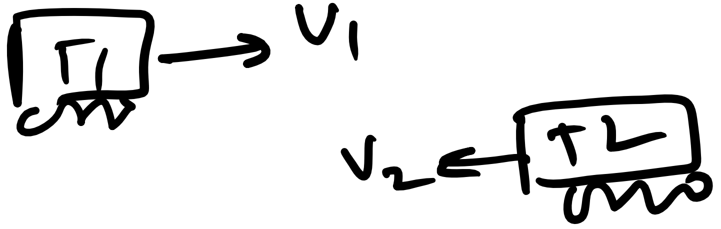
$$T = \frac{l_{T_1} + l_{T_2}}{V_{T_1} - V_{T_2}}$$



Time taken to cross each other

$$\boxed{T = \frac{D}{S}}$$

$$T = \frac{l_{T_1} + 0}{V_1 - V_2}$$



$$T = \frac{l_{T1} + l_{T2}}{v_1 + v_2}$$

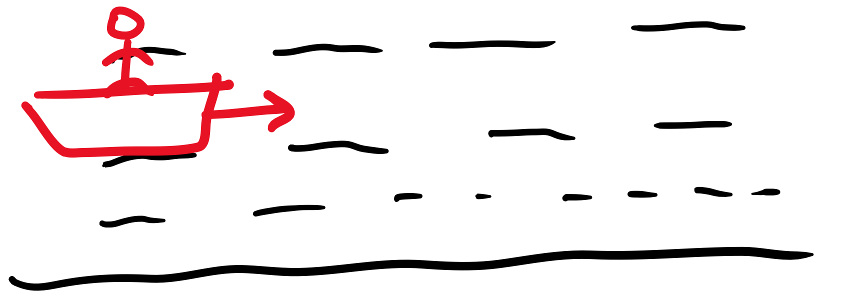
Q] From the time the front of a trains enters a platform, it takes 25s for the back of the train to leave the platform while train travelling at a constant speed of 54km/h .At the same speed , it takes 14s to pass a man running at 9km/h in the same direction as the train. What is the length of the train and that of the platform (in m) respectively?

(A)~~210~~,140
(B)~~162.5~~,187.5
(C)~~245~~,130
~~(D)~~175,200
[GATE 2018 ME]

① $\cancel{T}^{25s} = \dfrac{l_p + l_T}{\cancel{v_T} \rightarrow 54 \times 5/18 \ m/s}$

② $\cancel{T}^{14} = \dfrac{l_T + 0}{\cancel{v_T - v_p} \rightarrow (54-9) \times \frac{5}{18} \ m/s}$

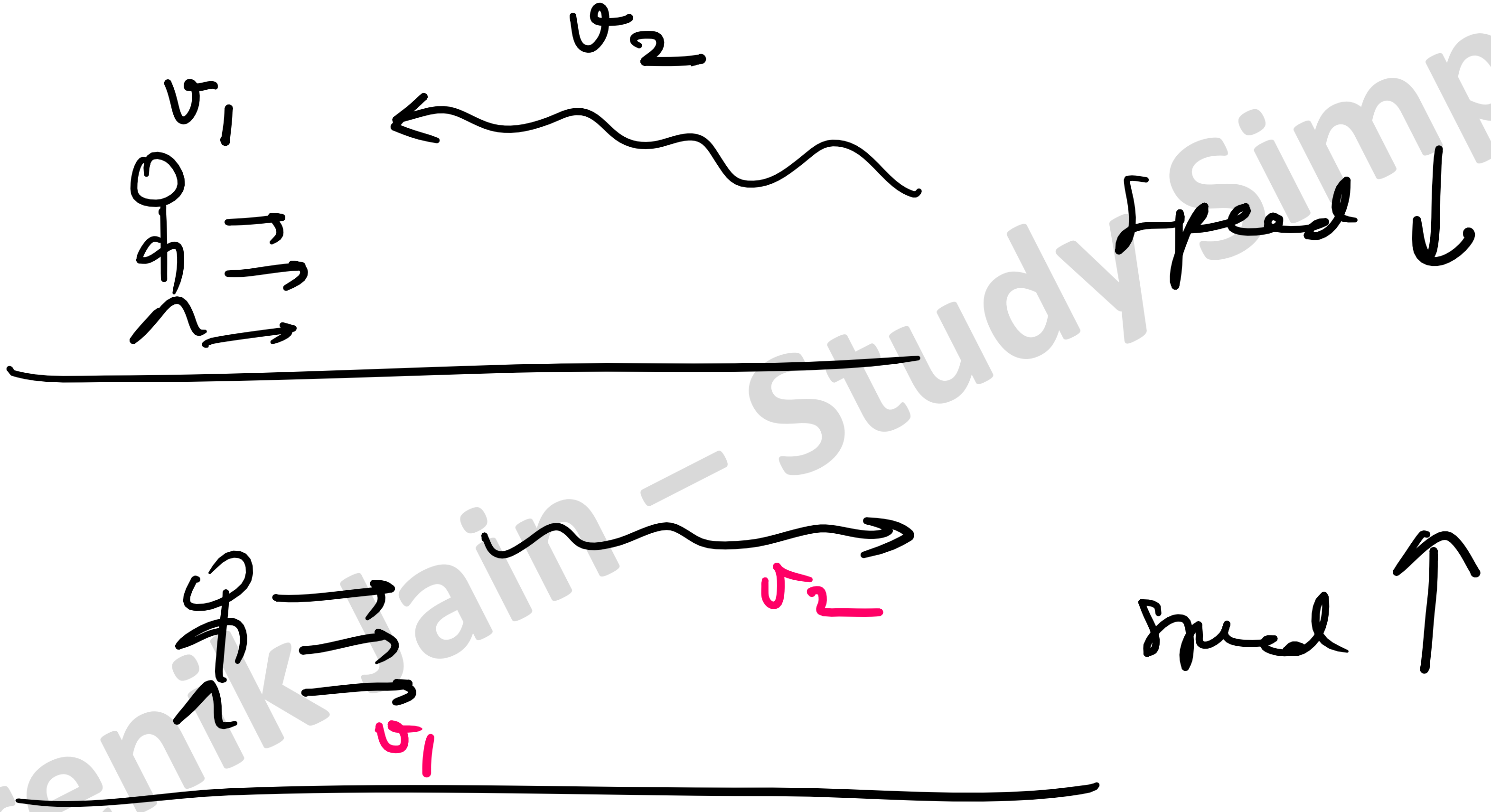
$\Rightarrow l_T = 175m$



B = Speed of boat in Still water

S = speed of water

$$\boxed{\begin{aligned} \text{Up} &= B - S \\ \text{Downstream} &= B + S \end{aligned}}$$



v2
v1
Speed ↓
v2
v1
Speed ↑

Q] A motorboat takes 2hr to travel a distance of 9km down the current and it takes 6hr to travel the same distance against the current. What is the speed of the boat in still water?

$\boxed{S = \frac{D}{T}}$

B

(1) Down the current (Speed ↑)

(Against current)

- Speed = B + S
- Speed = B − S

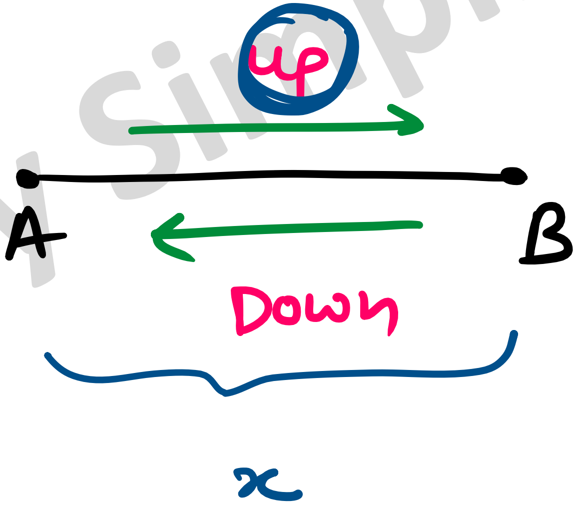
$B + S = \frac{9\,km}{2\,hr}$

$B - S = \frac{9\,km}{6\,hr}$

B = 3 km/hr, S = 1.5 km/hr

Q] A man can row at 5kmph in still water. If the river is running at 1kmph it takes him 75 minutes to row to a place and back. How far is the place?

$S = \frac{D}{T}$

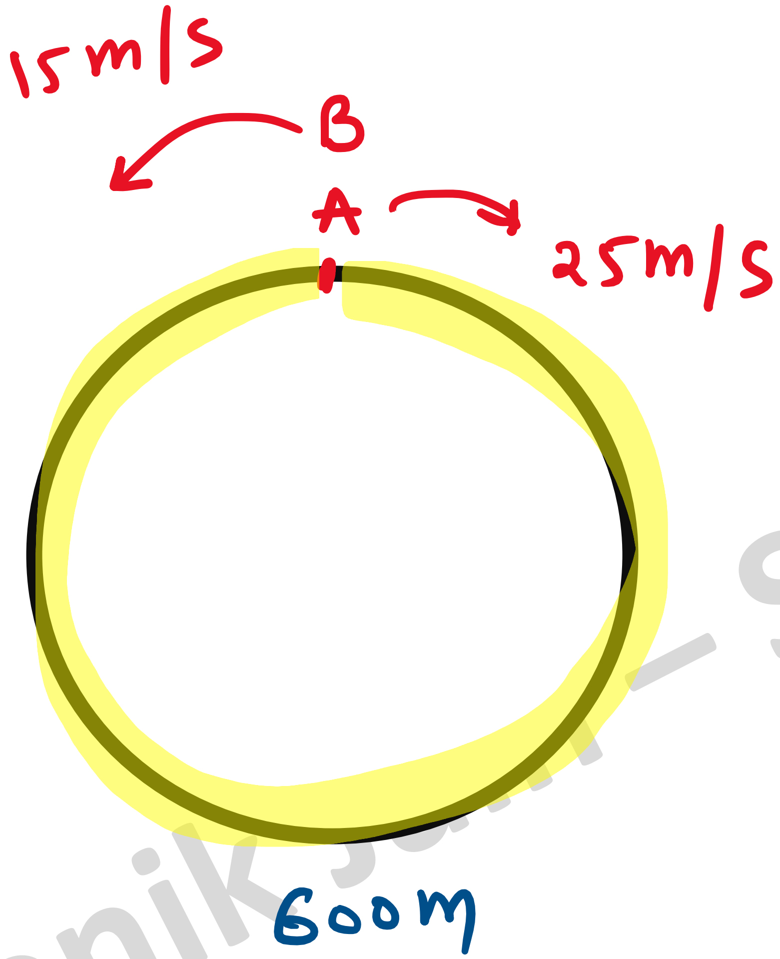


$$\boxed{\begin{aligned} B &= 5\ km/hr \\ S &= 1\ km/hr \end{aligned}}$$

$$T_{AB} + T_{BA} = 75\ min$$

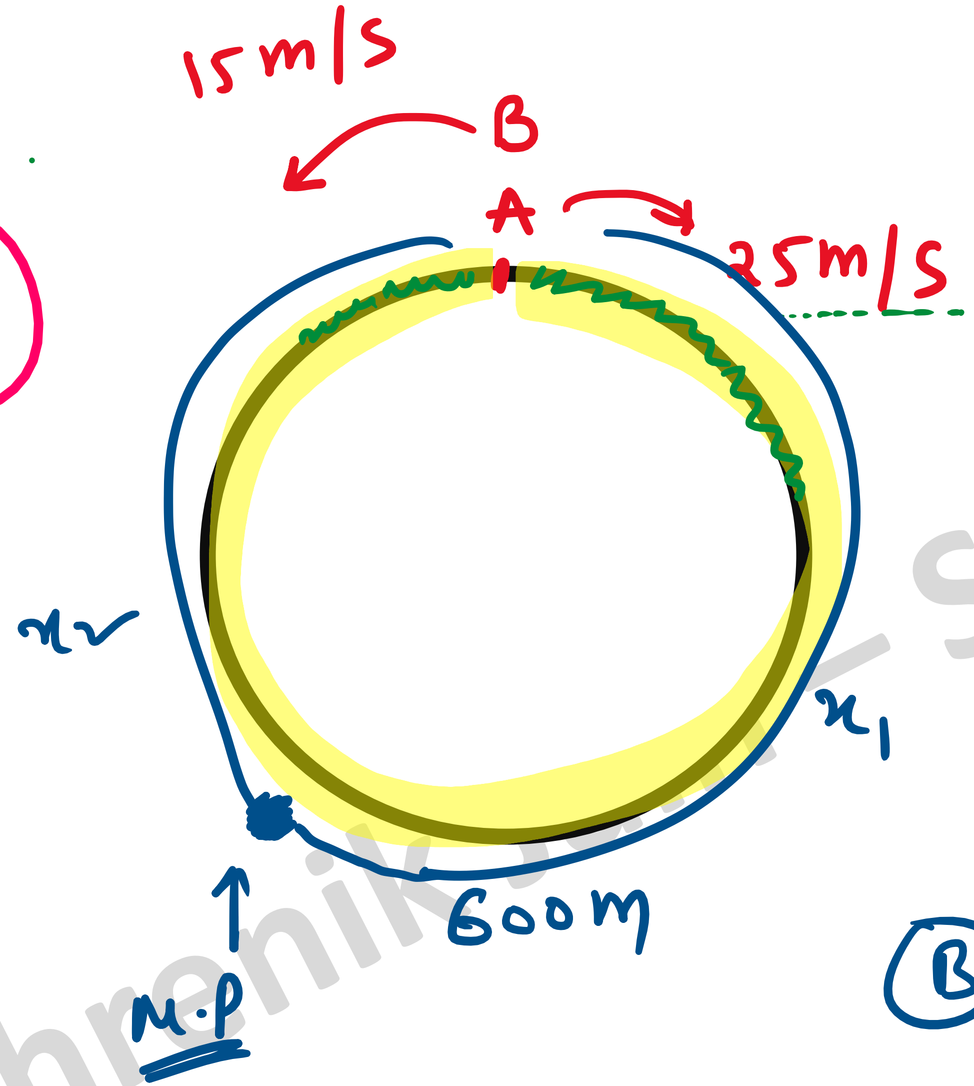
$$\frac{x}{\underset{5}{\cancel{B}} - \underset{1}{\cancel{S}}} + \frac{x}{\underset{5}{\cancel{B}} + \underset{1}{\cancel{S}}} = 75 \times \frac{1}{60}\ hr$$

$$\Rightarrow x = 3\ km$$



A & B are running then time taken to meet is ?

M1



A & B are running then time taken to meet is ?

Ⓐ $S = \frac{D}{T}$

$$\boxed{25\,m/s = \frac{x_1}{t}}$$

Ⓑ $15\,m/s = \frac{x_2}{t}$

$$\boxed{25\,m/s = \frac{x_1}{t}} \Rightarrow 25t = x_1$$

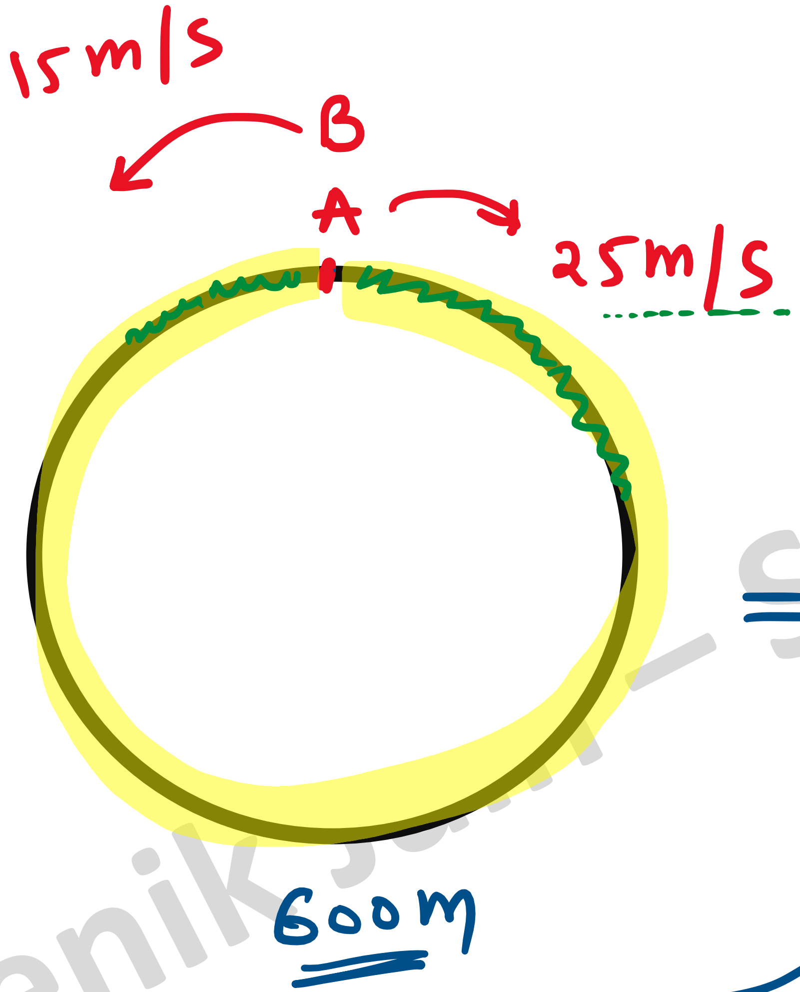
$$15\,m/s = \frac{x_2}{t} \Rightarrow 15t = x_2$$

$$t(25+15) = x_1 + x_2$$

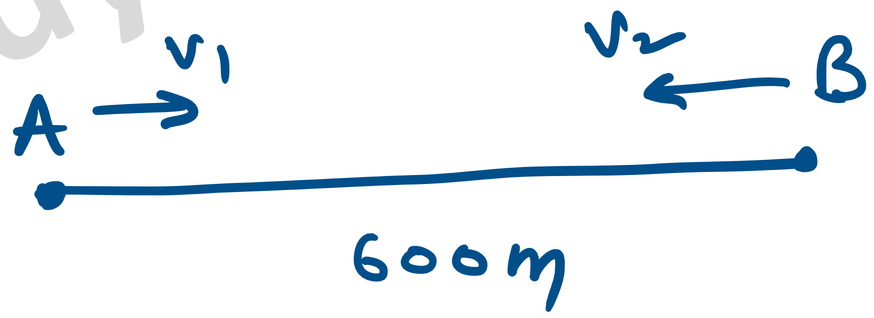
$$= 600$$

$$\therefore t = \frac{600}{40} = 15\,sec$$

M2



$\Rightarrow$



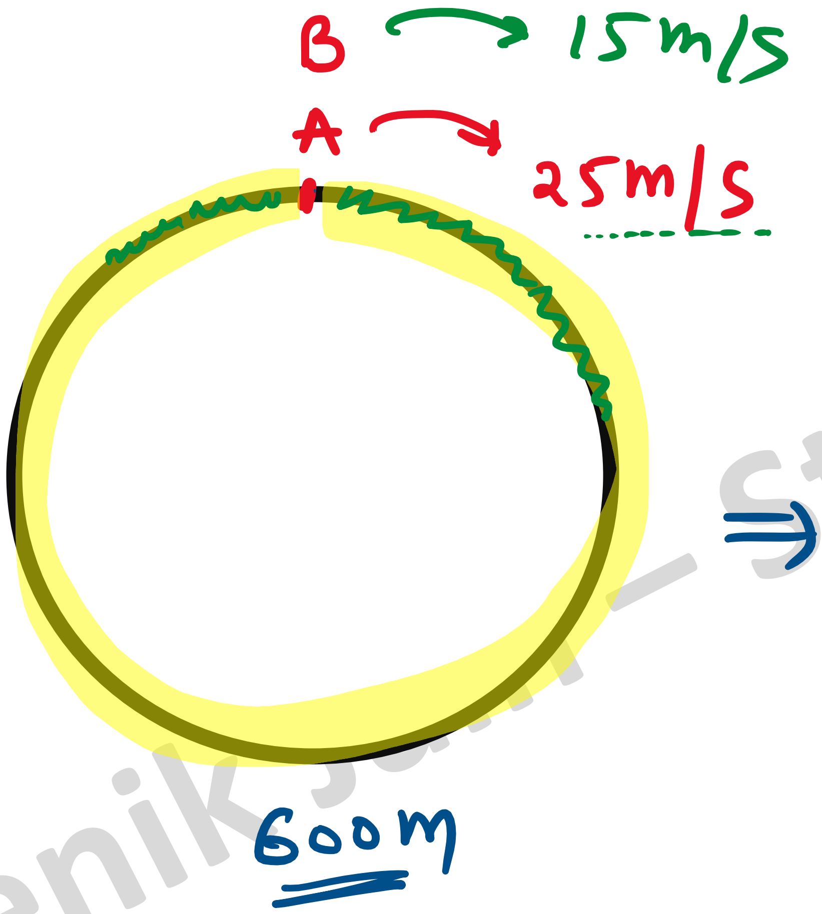
A & B are running then time taken to meet is ?

$S = D/T$

$$T = \frac{D}{S}$$

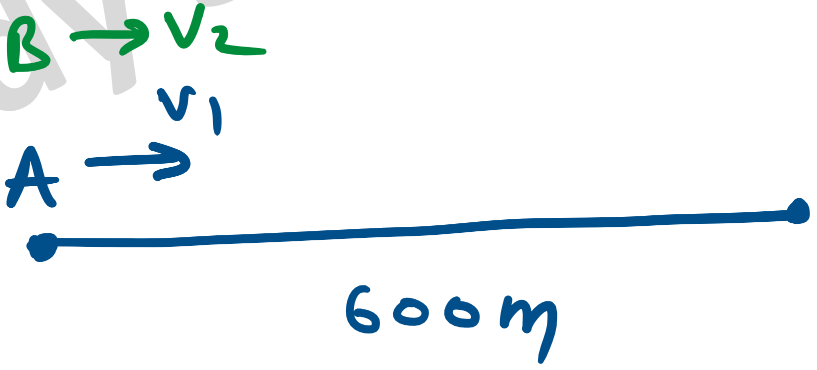
$$T = \frac{600}{v_1 + v_2} = \frac{600}{15+25} = 15\text{ sec}$$

M2



$\Rightarrow$

A & B are running then time taken to meet is ?



$$T = \frac{D}{S} = \frac{600m}{25-15}$$

$$= 60s$$